MOHAMED SATHAK
A J COLLEGE OF ENGINEERING
AN AUTONOMOUS INSTITUTION
SIRUSERI - CHENNAI

THIRUVIZHA

A SILVER JUBILEE CELEBRATION

SATHAK – A – THON 2.0 2026

- **Problem Statement ID –SI-SW**
- **Problem Statement Title- Agrisense A Smart Farming Assistant For diseases Prediction And Monitoring**
- **PS Category- Software+Hardware**
- **Team Name –InnovateXcrew**
- **Team Leader Name –D.RAGHUL**

AgriSense

PROBLEM STATEMENT

- Farmers lack accessible, real-time, and data-driven insights to combat crop diseases, optimize resource usage (water & fertilizer), and react to micro-climate changes.
- This reliance on guesswork and outdated traditional methods leads to massive crop yield losses, soil degradation, and severe economic strain on the farming community.
- **Production Loss**: According to the **FAO** (Food and Agriculture Organization of the United Nations), **up to 40%** of global crop yield is lost annually due to plant pests and diseases.
- **Economic Drain:** Plant diseases alone cost the global economy over **$220 Billion** each year.
- **Resource Inefficiency**: Agriculture accounts for **70%** of total global freshwater withdrawals, yet inefficient irrigation leads to massive water wastage.
- Soil Degradation: Blanket application of chemicals (without real-time NPK/TDS soil data) has degraded roughly **33%** of the Earth's soils, reducing long-term fertility.

Data from:

- FAO + International Plant Protection Convention (IPPC)
- FAO AQUASTAT database
- World Bank Water in Agriculture report
- Status of the World's Soil Resources (SWSR)

PROPOSED SOLUTION

1. Dual-Layer AI Disease Prediction
How it works: Farmers snap a photo of a leaf. The system uses a primary **Hugging Face ML Model** for instant detection.
The Safety Net: If the AI is uncertain, it auto-triggers a multimodal **Gemini Vision Layer** for a hyper-accurate second opinion, factoring in live local weather variables.

2. **IoT Soil Health Telemetry**
Continuous Monitoring: Low-cost, solar-ready ESP32 Soil Stations run continuously in the field.
Data to Action: They transmit live Soil Moisture and TDS (nutrient) readings to the cloud, automatically generating plain-text **Irrigation and Fertilizer advic**e before the crop suffers stress.

3. **Hyper-Localized & Accessible**
Native Language Voice Chatbot: AI diagnosis and treatment steps are spoken back to the farmer in their native language (Tamil/Hindi/English) via an integrated **Voice Assistan**t
Preventative Weather Forecasting: 7-day disease risk outlooks warn farmers of incoming humidity spikes, allowing them to spray **before** fungi spread.

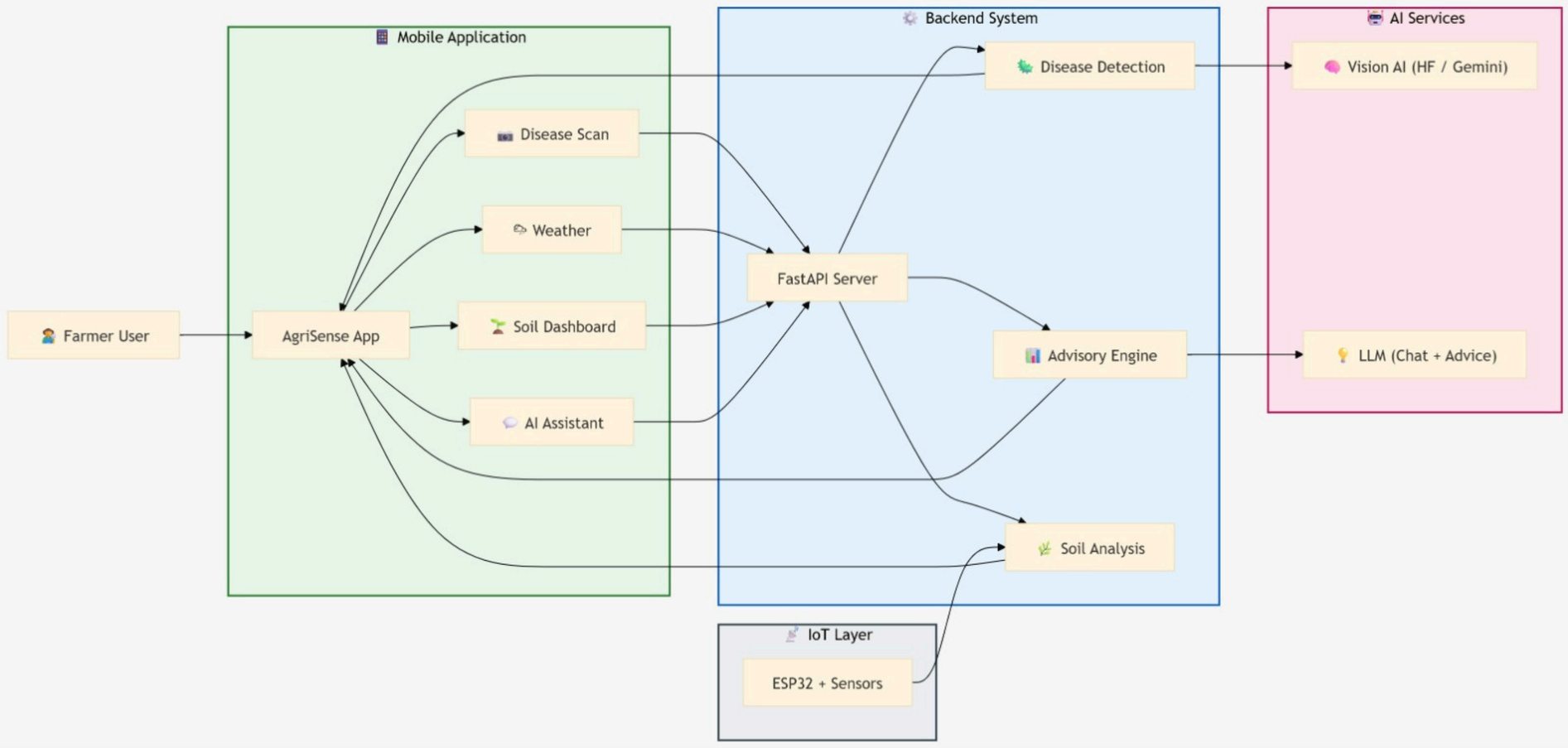
TECH STACK

Frontend (Mobile Application) :
Framework: React Native & Expo (Cross-platform iOS & Android)
Backend (API & Logic):
Framework: FastAPI (Python)
Hosting Context: Containerized via Docker and deployed on Hugging Face Spaces.

Artificial Intelligence & Computer Vision:
Machine Learning Inference: Hugging Face Serverless API (MobileNetV2 & Vision Transformers)
Visual Fallback & Chatbot: Google Gemini 2.5 Flash & Llama-3.3-70B-Instruct (via Hugging Face)
Computer Vision (Segmentation): Traditional OpenCV (HSV color masking, contour detection)

IoT & Hardware (Soil Station):
Microcontroller: ESP32 (Programmed in C++/Arduino IDE)
Sensors: DHT22 (Ambient Temp/Humidity), Analog Soil Moisture Sensor, Analog TDS Sensor.
Networking: ESP32 WiFi Stack sending JSON payloads via HTTP POST.

ARCHITECTURE / WORKFLOW



FEATURES AND INNOVATION

1. Custom IoT Hardware (AgriSense Soil Station)

A physical, field-deployable ESP32-based hardware edge device that runs on low-power deep-sleep cycles.
Outfitted with highly responsive Analog Soil Moisture sensors, TDS (Total Dissolved Solids) probes for accurate nutrient mapping, and DHT22 modules for localized ambient temperature and humidity tracking.

2. AI-Powered Disease Diagnostics

Instant leaf analysis via smartphone camera yielding highly accurate disease identification, confidence scores, and visual segmentation overlays.
Tabbed, easy-to-read treatment and prevention steps tailored exclusively to the detected disease.

3. Soil Health & Telemetry Dashboard
A mobile interface directly synced to the IoT Soil Station.
Auto-generation of specific, actionable watering and fertilization alerts directly derived from the live hardware telemetry (e.g., "High Salinity: Flush soil with fresh water").

4. Integrated Agricultural Calculators & Predictors
Fertilizer Algorithm: An on-device calculator determining exact bag quantities utilizing NPK values, plot acre size, and crop-specific demand multipliers.
7-Day Weather Risk Forecaster: Combines humidity and temperature into a normalized "Fungal Spread Risk" score, proactively alerting farmers before diseases strike.

FEASIBILITY AND VIABILITY

1. Technical Feasibility (Can we build & scale it?) :

Accessible Architecture: The mobile app is developed in React Native (Expo), allowing seamless cross-platform deployment to both Android and iOS devices from a single secure codebase.
Hardware Modularity: Using off-the-shelf ESP32 microcontrollers with DHT22 and Analog TDS/Moisture sensors means the physical Soil Stations are easily sourceable, easy to mass-produce, and easy to replace.
Reliable AI Infrastructure: Instead of forcing heavy machine learning models onto the farmer's low-end smartphone, all heavy-lifting is containerized globally on FastAPI and the Hugging Face Serverless Inference API, requiring only a basic internet connection from the user.

2. Economic Viability (Is it sustainable & affordable?) :

Low Barrier to Entry: By stripping out expensive enterprise data-loggers and replacing them with a custom $15-$25 IoT module, precision agriculture becomes genuinely affordable for the average smallholder farmer.
Operational Cost Efficiency: The backend leveages Open-Source models and serverless scaling, ensuring cloud architecture costs scale linearly instead of exponentially.
Long-Term Value Creation: By using the exact right amount of fertilizer (via the algorithm) and spraying fungicide ***only*** when the 7-day risk forecaster flags it, farmers massively reduce chemical waste, ensuring a fast Return on Investment (ROI) while regenerating soil health.

RESEARCH AND REFERENCES

S.No	Existing System & Reference	Description	Drawbacks	How AgriSense Overcomes
1	**AI Plant Disease DetectionS. Mohanty et al., (2016), IEEE**	Deep learning (CNN) models trained on leaf images (PlantVillage dataset)	Works only on lab datasets No real-world adaptation No fallback for low confidence No environmental context	Dual AI system (ML + Gemini) Confidence-based validation Uses weather + season data Higher real-world accuracy
2	**IoT Smart Irrigation SystemA. Kumar et al., (2019), Elsevier**	Uses soil moisture sensors to automate irrigation	Only threshold-based No nutrient analysis No intelligence or prediction No integration with disease detection	Uses Moisture + TDS + Climate Smart irrigation advice Fertilizer recommendations Integrated with AI disease system
3	**Weather Advisory SystemsIndian Meteorological Department (IMD)**	Provides weather forecasts and alerts	Not crop-specific No disease linkage No field-level data No actionable insights	Crop-specific risk analysis Disease prediction using weather Micro-climate integration AI-based recommendations
4	**Traditional Farming Methods(General practice, FAO reports – Food and Agriculture Organization)**	Manual observation and experience-based decisions	Guesswork decisions Overuse of water & fertilizers Late disease detection Soil degradation	Data-driven decisions Real-time monitoring (IoT) Early disease detection Optimized resource usage

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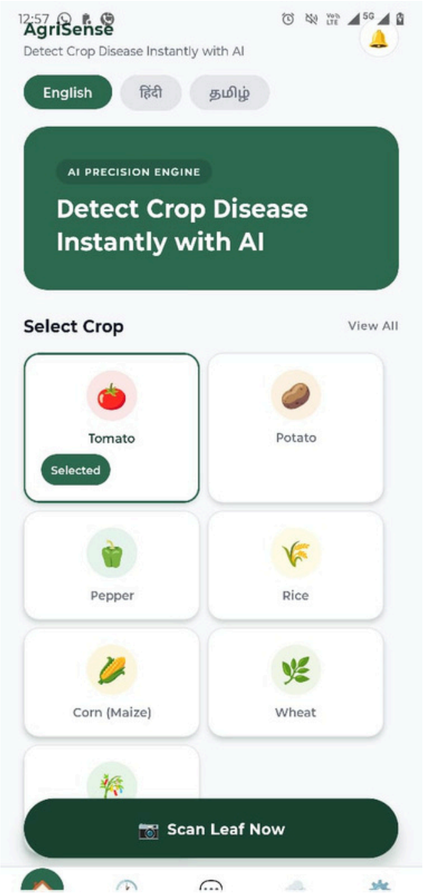
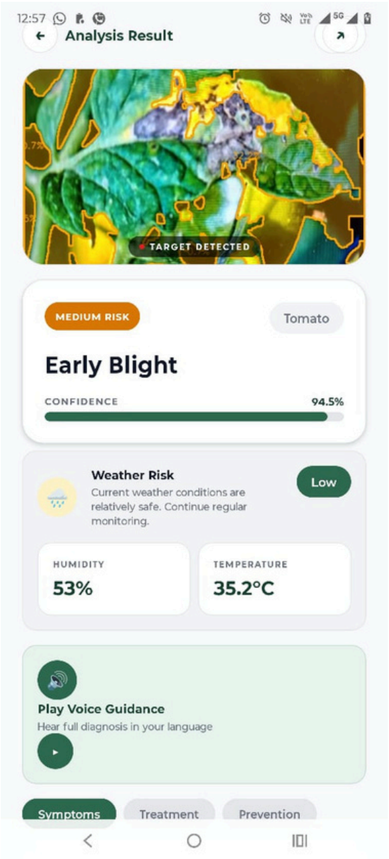
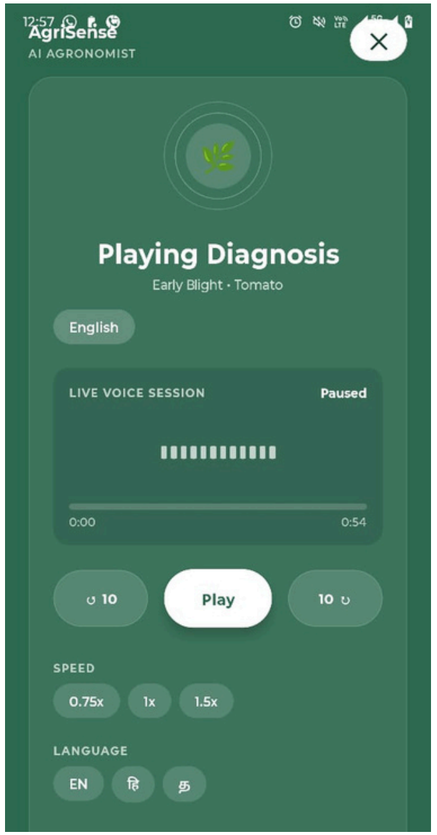
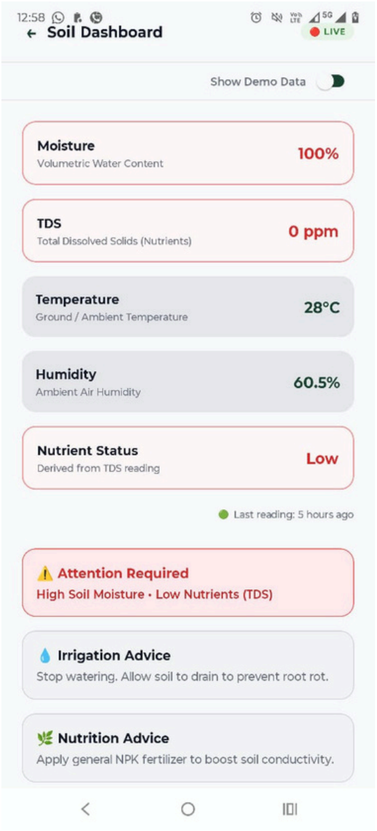
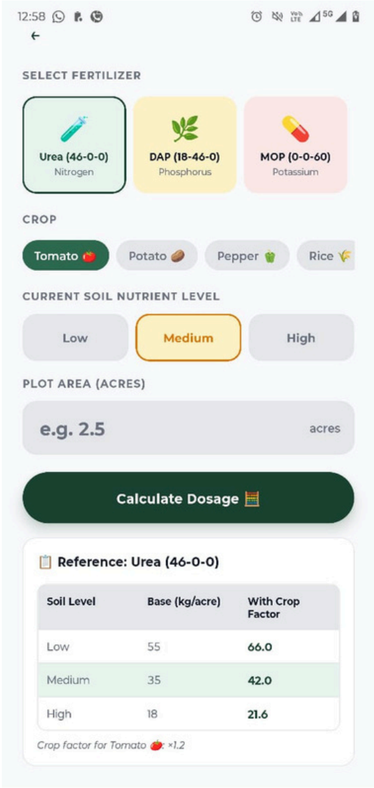
IMPLEMENTATION / DEMO



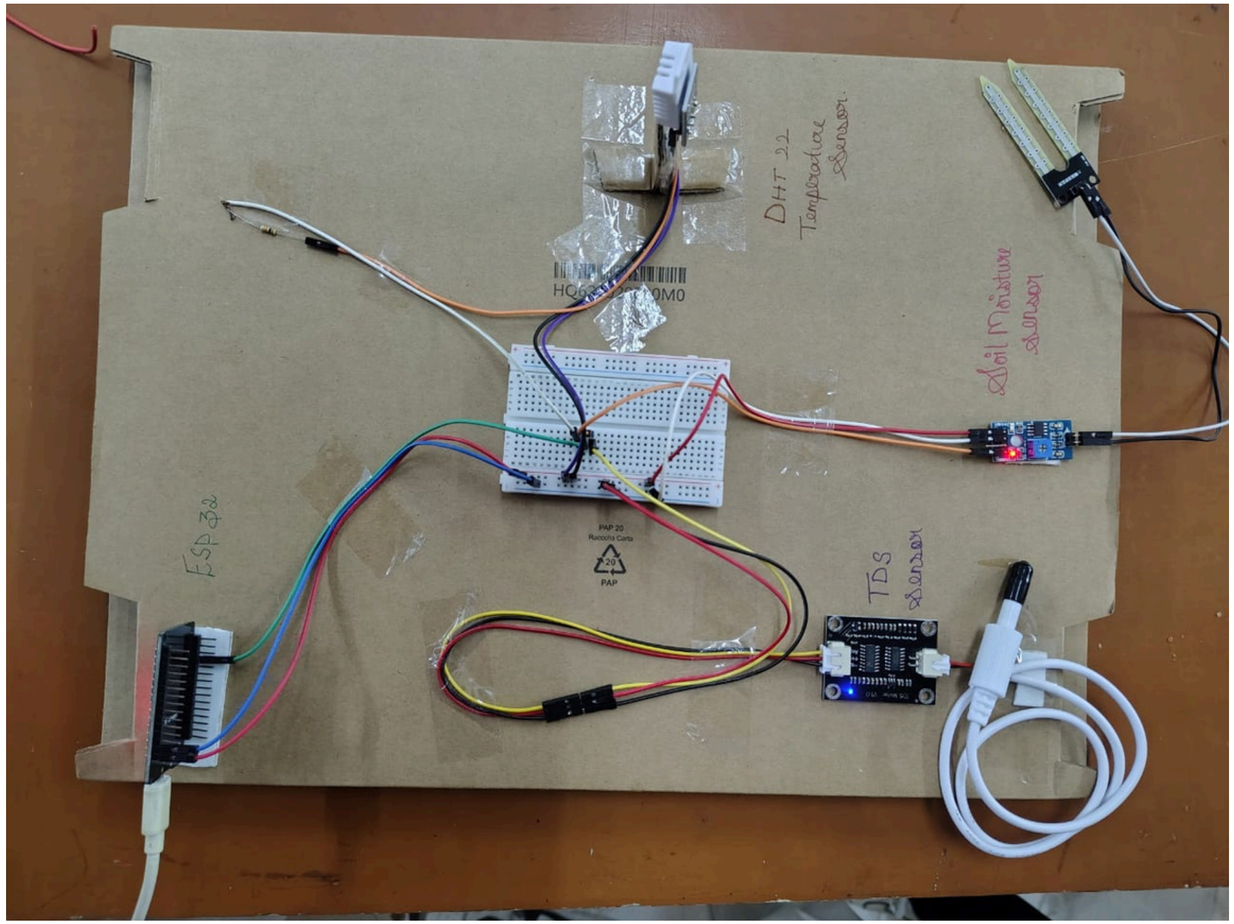








Hardware module



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